

LANDroid Rebuild Plan

Side-by-Side Architectural Review

Codex (OpenAI) and Claude (Anthropic, Opus 4.7)

Prepared 2026-05-21 - branch: codex/rebuild-plan-review-claude-2026-05-21

LANDroid Rebuild Plan - Side-by-Side Architectural Review

Document purpose: This PDF bundles two independent architectural reviews of docs/rebuild-plan.md so the next reviewer (Codex, in this case) can see both perspectives at once and a structured comparison between them.

Contents:

- **Section 1.** Codex's review (verbatim, as provided to Claude).
- **Section 2.** Claude's first review (original 15-section deep review against the rebuild plan).
- **Section 3.** Claude's second review (fresh redo, deliberately not anchored to Section 2 until written).
- **Section 4.** Claude's side-by-side comparison of Codex vs Claude (Reviews 1 and 2 combined).
- **Section 5.** Recommended next steps for amending the plan and running follow-up audits.

How to use this document:

The reviews are independent. Where Codex and Claude agree, treat that as a strong signal. Where they disagree, the comparison section in Section 4 names the disagreement and which side I (Claude) think is right and why.

Branch: codex/rebuild-plan-review-claude-2026-05-21

Reviewers:

- Codex (OpenAI) - Section 1.
- Claude (Anthropic, Opus 4.7) - Sections 2, 3, 4, 5.

Date assembled: 2026-05-21.

Source plan reviewed: docs/rebuild-plan.md at the date above.

Not implementation: No code changes were made. This is a planning artifact only.

Table of Contents

Section 1 - Codex's Review

what Codex thinks of the rebuild plan

Section 2 - Claude's Review 1 (Original)

first deep review with parallel external research

Section 3 - Claude's Review 2 (Fresh Redo)

second pass, deliberately not anchored to Review 1

Section 4 - Side-By-Side Comparison

agreements, disagreements, consolidated amendment list

Section 5 - Recommended Next Steps

how to amend the plan and run follow-up audits

Section 1 - Codex's Review

Source: provided by the user as Codex's review of docs/rebuild-plan.md . Reproduced here verbatim.

1. Executive Recommendation

Conditional go. The current direction in docs/rebuild-plan.md (line 70) is broadly right: Project record graph as domain truth, documents as first-class entities, and typed approval/action records over the top. That aligns with serious legal/title systems: legal DMS products center work around matters/projects, documents, versions, permissions, and search; eDiscovery systems package native files, text, metadata, and load files; land/title tools tie documents directly to tracts, ownership, leases, and title interpretations.

But the plan is not yet specific enough to start rebuild work. The missing professional layer is an evidence-grade document architecture: immutable originals, hashes, document versions, extraction runs, page/span anchors, search indexes, packet manifests, and chain-of-custody fields. Without that, LANDroid could build a nice graph and still fail at the thing title work needs most: proving exactly where an answer came from.

My recommendation: keep the plan, amend it before implementation, and promote the document/evidence storage contract ahead of any runsheet wizard or AI Q&A build.

2. What The Current Plan Gets Right

- Rejecting pure event sourcing is correct. A title project is not naturally an event log; it is a living project file with records, source documents, citations, review states, and derived views.
- Documents outside events is correct. PDFs should never be buried inside action history.
- Phase 0 inventory and golden masters are non-negotiable. The app already has real workflows worth preserving.
- SourceAttestation is a strong concept. Starting from a prior title opinion is real title practice and should be modeled as provenance, not faked as a patent root.
- Runsheet package import/export is correctly treated as a deterministic template workflow, not an AI free-for-all.
- AI answer contract is directionally right: answer, citations, limits, open issues, next action.
- Deferring federal/private math is right. The current Texas-only boundary in PROJECT_CONTEXT.md (line 13) should not be blurred.

3. What The Plan Gets Wrong Or Under-Specifies

- The Document Vault section is too thin. It says stable Document and DocumentLink, but it does not define DocumentVersion, immutable original file, file hash, MIME/magic validation, derivative OCR PDF, extracted

text, page images, text spans, extraction engine/version, confidence, or audit trail.

- The .landroid packaging model is still vague. "Hybrid package/snapshot plus action records" is right conceptually, but the plan needs a concrete package layout before format work starts.
- The action layer is slightly too late. Phase 3 already needs typed ActionPlan dry runs for import review. Phase 4 can make action records canonical, but Phase 3 cannot be built cleanly without the action-plan schema.
- The AI contract is underspecified technically. "Cited sources" is not enough. LANDroid needs a verifier that refuses answers unless each material claim maps to a stored SourceCitation or deterministic calculation result.
- The backend boundary needs a decision gate. Local-first is right for now, but OCR jobs, cloud AI, hosted persistence, and object storage need adapter boundaries now so a backend can be added without rewriting storage again.

4. Better Architecture

Use the current plan, but tighten it into five layers:

- Project Package: one portable project folder/zip with schema version, project DB/snapshot, vault files, manifests, checksums, and optional rebuilt indexes.
- Record Graph: normalized project records: parties, aliases, tracts, documents, instruments, leases, units, wells, source attestations, curative issues.
- Evidence Vault: immutable source files plus derived files, text extraction, OCR results, page/span anchors, hashes, review status, and document versions.
- Action Layer: typed ActionPlan previews, approvals, ActionRecords, rollback boundaries, and durable audit records.
- Projections: Desk Map, Runsheet, Leasehold, Documents, Owner index, Packet export, and AI context.

This is closer to how NetDocuments describes a secure single system of record with versioning and matter/project workspaces, how iManage positions document/email management around protected knowledge work, and how eDiscovery platforms like Relativity import metadata, natives, and extracted text through load files. Sources: NetDocuments, iManage, Relativity import docs.

5. Document Storage Recommendation

Short term: keep IndexedDB/Dexie as the runtime store because that is the current app architecture.

Target: do not rely on browser IndexedDB as the only durable home for large title projects. MDN notes IndexedDB is useful for large offline data, but browser storage can still be wiped by user action, private browsing, quota pressure, corruption, or incompatible changes, and it has no native full-text search equivalent. Source: MDN IndexedDB.

Best target package:

```

ProjectName.landroidpkg/
  project.json or project.sqlite
  manifest.json
  manifest.csv
  checksums-sha256.txt
  documents/
    originals/
      <documentId>_<sha256>.pdf
    derivatives/
      <documentId>_ocr.pdf
      <documentId>_page-001.png
  text/
    <documentId>.txt
    <documentId>.json
  exports/
    Runsheet.xlsx
    TitleDocuments/
  indexes/
    rebuildable-search-index-files

```

A zipped .landroid package is fine for transfer/export. For active work, a folder/package is safer than one huge JSON blob. BagIt-style manifests are worth copying because they use payload files plus checksum manifests for integrity. Source: APTrust BagIt FAQ.

Cloud object storage should be adapter-ready but deferred. Later, S3/R2/GCS can store originals/ and derivatives/; the LANDroid record graph remains the source of truth.

6. OCR/Text Extraction Recommendation

Use a two-track pipeline:

- For selectable PDFs: extract text, blocks, words, and coordinates with PyMuPDF.
- For scanned PDFs: run OCRmyPDF/Tesseract locally first, preserving the original and creating an OCR derivative.

OCRmyPDF adds an OCR layer back onto the original PDF-derived page image and is limited by Tesseract; PyMuPDF can extract word/block coordinates, but reading order is heuristic. Sources: OCRmyPDF, Tesseract, PyMuPDF text extraction.

Required records:

```

ExtractionRun
  documentId
  documentVersionId
  engine
  engineVersion
  startedAt
  completedAt
  status

```

```
confidenceSummary
parameters
outputArtifactIds
```

```
SourceCitation
documentId
documentVersionId
extractionRunId
pageNumber
bbox / polygon / textSpan
quotedText
quotedTextHash
citationType
```

Cloud OCR is worth testing for poor scans or handwriting. As of this review, official basic OCR pricing is roughly low per page: Google Document AI Enterprise OCR lists \$1.50 per 1,000 pages for the first 5M pages, AWS Textract Detect Document Text examples show \$0.0015/page for the first 1M pages, and Azure Document Intelligence Read is in the same general tier but should be checked in the Azure calculator because the page renders region/account-specific prices.

7. AI/RAG/Search Recommendation

Use hybrid retrieval, not vector search alone.

For LANDroid, exact search matters as much as semantic search: instrument numbers, volume/page, county names, legal descriptions, party aliases, tract codes, lease names, and well names are often exact-match facts. The best architecture is:

- SQLite FTS5/BM25 or similar for exact/keyword search.
- Vector embeddings for natural-language questions.
- Graph traversal for "where does this party appear?" and chain questions.
- Deterministic math tools for ownership and leasehold calculations.
- A citation verifier before answers are displayed.

SQLite FTS5 provides full-text search and BM25 ranking; pgvector explicitly supports combining vectors with Postgres full-text search; Qdrant and Meilisearch both support hybrid search patterns.

OpenAI File Search is useful as a prototype or hosted option, but it should not be LANDroid's canonical evidence layer because LANDroid needs its own citation, versioning, deletion, and project-package guarantees. OpenAI's docs note vector stores auto-chunk/embed/index files and that file removal can be eventually consistent.

8. Attorney Packet/Export Recommendation

The current Excel plus TitleDocuments/ direction is right, but make it more professional:

```

AttorneyPacket/
  README.txt
  Runsheet.xlsx
  TitleDocuments/
    RF-000001_2020-05-01_Lease_Smith-to-ABC.pdf
  manifest.csv
  manifest.json
  loadfile.csv
  source-citations.csv
  unresolved-issues.csv
  checksums-sha256.txt
  ocr-text/
    RF-000001.txt

```

The workbook should use relative hyperlinks into TitleDocuments/. manifest.csv should be human-readable. manifest.json should be machine-complete. loadfile.csv should mirror eDiscovery practice: DocID, native path, text path, document date, parties, instrument number, volume/page, tract, source type, hash.

EDRM explains that load files connect images, full text, metadata, and natives so they work together; Relativity similarly imports load files containing metadata, native files, and extracted text.

9. Local Tools To Install On The Mac

Free/open-source, high value:

- ocrmypdf, tesseract, tesseract-lang, ghostscript: local OCR.
- qpdf, pikepdf: PDF validation, linearization, repair, safe transformations.
- poppler: pdftotext, pdftoppm, page rendering.
- mupdf / mutool: fast PDF inspection/rendering.
- exiftool: metadata inspection.
- sqlite, duckdb: local index/analysis experiments.
- Python packages: pymupdf, pdfplumber, pikepdf, ocrmypdf, openpyxl, duckdb.
- Optional later: Postgres.app plus pgvector if backend/RAG experiments move beyond browser storage.

No need to install these in the app yet. Treat them as development and pipeline tooling first.

10. Paid Services/Tools Worth Considering

- Enverus Courthouse: high fit, likely paid/enterprise. It claims AI over courthouse/title records with page/paragraph citations and runsheet workflows. Useful benchmark or data source, not a LANDroid replacement.
- Tracts: title automation benchmark; it markets document interpretation placed into chain/calculation workflows. Good competitive reference.

- Quorum / On Demand Land / My Quorum Land: enterprise land system pattern. Strong confirmation that title documents, leases, tracts, ownership, GIS, OCR, and auditability belong together. Likely too heavy/expensive for now.
- NetDocuments / iManage: legal DMS benchmark for document versioning, secure matter workspaces, audit, permissions, and AI over governed documents. Probably not LANDroid's internal storage layer.
- Google Document AI / AWS Textract / Azure Document Intelligence: paid OCR/extraction fallback for difficult scans.
- ABBYY FineReader PDF or ABBYY Vantage: FineReader may be useful locally; Vantage is enterprise IDP and likely too much unless local OCR quality is inadequate.
- OpenAI embeddings: cheap enough for indexed OCR text experiments; OpenAI lists text-embedding-3-small at \$0.02 per 1M tokens.

11. Open-Source Tools Worth Considering

- OCR/text: OCRmyPDF, Tesseract, PyMuPDF, pdfplumber, pikepdf, qpdf, Poppler, PDF.js.
- Local search/indexing: SQLite FTS5, SQLite vec1/sqlite-vec, DuckDB for analytics.
- Backend search later: Postgres + pgvector, Qdrant, Meilisearch.
- RAG orchestration: LlamaIndex or LangChain only if they save work; avoid letting them own LANDroid's citation model.
- Packaging/integrity: BagIt-style manifests and SHA-256 checksum validation.

12. Specific Edits For docs/rebuild-plan.md

- Add a new "Evidence Vault Contract" before or inside the current Document Vault section at docs/rebuild-plan.md (line 209). It should define immutable originals, document versions, content hashes, derivatives, extraction runs, page/text/span anchors, and citation verification.
- Add these target records to the Phase 1 list at docs/rebuild-plan.md (line 411):

```
DocumentVersion
VaultObject
ExtractionRun
DocumentPage
DocumentTextSpan
CitationAnchor
Packet
PacketItem
PacketExport
SearchIndexEntry
```

- Add a concrete .landroid package specification near docs/rebuild-plan.md (line 32): one package/folder/zip with project.json or project.sqlite, documents/originals, documents/derivatives, text sidecars, manifests, checksums, and rebuildable indexes.

- Revise Phase 3/4: Phase 3 should create typed ActionPlan dry runs for imports; Phase 4 should make durable ActionRecords canonical across more workflows.
- Add a "Search And AI Evidence Contract": hybrid search, deterministic graph/math tools, citation verifier, unresolved answer behavior, and explicit refusal to cite generated summaries as sources unless they point back to original pages/spans.
- Add a "Backend Decision Gate": local-first remains source of truth now; backend is introduced only for hosted sync, object storage, OCR jobs, multi-user collaboration, or server-held AI/RAG.

13. Edits For Other Source-Of-Truth Docs

- ROADMAP.md (line 8): promote "Evidence Vault/package spec" into Now, before packet export and AI document query.
- IDEAS.md (line 70): move tokenized DMS-style search from Parked to Promising; search is core, not a side idea.
- ARCHITECTURE.md (line 149): add "Current source of truth vs target source of truth." Today Desk Map is source of truth; target is project record graph with Desk Map as projection.
- TESTING.md (line 63): add tests for document hash preservation, package round-trip, packet manifest determinism, OCR citation anchors, search retrieval fixtures, and answer citation verification.
- SECURITY.md (line 104): expand OCR planning into cloud OCR policy, chain-of-custody, document hashes, external provider disclosure, and retention/deletion rules.

14. Risks And Migration Costs

- Wrong storage choice: storing PDFs in one JSON file or only in browser storage will be expensive to unwind.
- Weak citations: retrofitting page/span citation anchors after AI Q&A ships will be painful.
- Vector-only search: exact title terms will be missed, creating false confidence.
- No document versioning: title opinions, revised packets, OCR derivatives, and attorney edits will blur together.
- No hashes: LANDroid cannot prove a packet file matches the reviewed source.
- Premature backend: cloud sync/object storage before local package semantics are stable will multiply migrations.
- Automatic party dedupe: merging owners without review can corrupt title facts across a 60,000-acre project.
- Cloud OCR/AI without policy: sensitive title docs and owner info could leave the machine without an explicit user decision.

15. Final Go/No-Go Recommendation

- Go on Phase 0 inventory and planning edits.

- No-go on starting the rebuild implementation or full runsheet walkthrough wizard until docs/rebuild-plan.md (line 1) is amended with the evidence vault, package format, search/index, OCR/citation, and backend decision contracts.
- No-go on making a backend the canonical source of truth right now. Build local-first with backend-ready adapters.
- Go on the architecture itself after amendments: project record graph + evidence-grade document vault + typed action layer is the right professional direction.

Section 2 - Claude's Review 1 (Original 15-Section Deep Review)

Reviewer: Claude (Anthropic, Opus 4.7). Date: 2026-05-21. Method: deep read of all repo source-of-truth docs plus parallel external research across legal DMS, eDiscovery, title/land vendors, OCR, RAG, and local-first storage.

1. Executive Recommendation

Adopt the plan with three structural amendments, not a rewrite.

The project record graph + document vault + action layer is the right model. It matches how Tracts.co shapes the same problem and is more rigorous than every agreement-centric upstream vendor I surveyed (Quorum, P2 iLandMan, W Energy, Pandell, LandVantage, Landboss). The plan correctly rejects pure event sourcing, correctly keeps documents out of events, correctly puts Phase 0 (inventory + golden masters) before everything, and correctly defers math expansion.

But three load-bearing things in the current plan are wrong or under-specified, and if you don't fix them before you start building, the rebuild will collapse on Raven Forest:

- 1. The `.landroid` storage format must change.** Today, the workspace is stored as `data: string` - a single JSON blob inside one IndexedDB row, autosaved every change. A 60,000-acre project with thousands of PDFs, OCR text, units, leases, and ownership nodes will produce a multi-megabyte string that is parsed/re-stringified on every autosave. The plan never names this and it should be the first thing fixed.
- 2. The plan defers OCR but promises cited AI answers.** You cannot honestly cite document evidence without text-with-bounding-boxes. The "AI Answer Contract" is uncashable until the OCR/citation pipeline exists. Either the AI Q&A is *structured-record-only* until OCR lands, or OCR moves earlier.
- 3. The side stores are still fragmented.** `ownerDocs`, `mapAssets`, `researchImports`, and `documents` all exist as separate tables, each with its own attachment shape. Audit finding P2-DOCS-1 calls this out. The plan's "Document Vault" depends on unifying these, but the plan doesn't make that unification a phase gate. It should be Phase 2's first task, ahead of new vault features.

Plus four additions: an `OpinionDraft` projection, a `LeaseObligation` calendar, an append-only `audit_events` table, and an attorney packet that's parallel-compatible with Concordance/Opticon load files. None of these are big; all are absent from the current plan.

Verdict: green-light Phase 0 as written. Before Phase 2 starts, address the three amendments. Most of the rest of the plan stands.

2. What The Current Plan Gets Right

- **Project record graph, not event log.** Pure event sourcing for a domain like this becomes a graveyard of replays. CQRS-lite (records as truth, actions as audit) matches how every serious vendor actually models. The plan reaches this conclusion explicitly.
- **Phase 0 = current behavior inventory + golden masters.** This is the single most professional decision in the document.
- **SourceAttestation as the title-chain root primitive.** I could not find this concept named in any competitor. Quorum, P2 iLandMan, Enverus, and Landboss all silently start from "current owner of record" with no first-class artifact saying *"we are starting from a 2018 DOTO, not from patent."* That's a real competitive differentiator and a genuine professional improvement. Keep it.
- **Documents as stable first-class entities with many polymorphic links.** Already 80% built.
- **Dual decimal + rational fraction display as a contract.** Title work breaks when display formats drift. Locking this is correct.
- **Rejecting pure event-sourced .landroid , rejecting PDF blobs in events.** Both correct.
- **Title-opinion-as-root workflow.** Maps directly to how landmen actually start - almost no chain is examined back to sovereignty cold.
- **Approval-gated AI with typed previews and undo.** Already implemented.
- **Federal/private kept as reference-only until Phase 2 math gate.** Correct.
- **Math engine preserved.** LANDMAN-MATH-REFERENCE.md is one of the most rigorous landman-facing math docs in this category. Don't touch it during rebuild.

3. What The Plan Gets Wrong Or Under-Specifies

3a. The workspace storage format is the elephant in the room (severity: very high). The current .landroid v8 file is JSON. The autosaved workspace lives as `workspaces.data: string`. Every autosave parses and re-stringifies the entire workspace. Document blobs are correctly external in `documents`, but the *graph* - nodes, deskmaps, leasehold unit, leases, ORRIs, assignments, transfer-order entries, instrumentTypes, owners, research, maps - accumulates into one JSON string. For Raven Forest scale (~60,000 acres, thousands of instruments, possibly 5,000+ ownership nodes, dozens of leases, multiple units), realistic 10-30 MB workspace JSON. Stringify + IndexedDB write on every autosave at that size = visible UI stalls on save, autosave write amplification, and a real risk of corruption under partial-write failures. **This is the single highest-leverage change in this review.**

3b. The OCR/AI citation contract is uncashable as written (severity: high). The "AI Answer Contract" requires cited sources. But the plan defers OCR engine implementation. The contradiction: AI cannot cite documents it has not parsed. Two honest paths: (1) defer the document-citing form of AI; (2) bring a real OCR pipeline forward via `ocrmypdf` + Apple Vision plugin.

3c. Side-store fragmentation blocks the vault promise (severity: high). Audit finding P2-DOCS-1 names this. Without unification: attorney packet export keeps missing owner-side and research-side documents; AI Q&A retrieval surface is incomplete by definition; the "one source of truth" goal in the plan's summary is structurally false. Phase 2 should start with: collapse `ownerDocs` / `mapAssets` (PDF subset) / `researchImports` (file subset) into `documents`.

3d. The plan has no opinion-shaped output projection (severity: medium-high). Every competitor produces a title opinion deliverable. LANDroid already has every input record needed: nodes, leases, NPRIs, ORRIs, curative issues, source attestations. What's missing is an `OpinionDraft` projection that renders the records into the canonical opinion structure (Caption -> Instruments Examined -> Tract Description -> Surface Ownership -> Mineral Ownership Table -> Leasehold Ownership -> Comments -> Requirements/Curative -> Exceptions/Qualifications).

3e. No obligation/calendar layer (severity: medium). Every upstream land vendor I checked has one: lease expiration, Pugh, shut-in, rentals, options. For a 60,000-acre project with multiple operators and units, missing a deadline is the failure mode that costs real money.

3f. Attorney packet is not eDiscovery-compatible (severity: medium). The current human-readable package is great, but every eDiscovery platform I surveyed reads a 30-year-old standard: Concordance `.DAT` + Opticon `.OPT` / `.LFP` with BegBates / EndBates / BegAttach / EndAttach / Custodian / MD5 / SHA256 . Adding a parallel sidecar costs almost nothing.

3g. No chain-of-custody / hash-chained audit log (severity: medium). `ActionRecord` is listed but durability and tamper-evidence are missing. The audit table should be append-only with hash chaining; the export manifest should carry a Merkle root.

3h. Project boundary is good; cross-project party identity is the next moat (severity: low). Note an `external_party_id` field on `Party` now so a later cross-project canonical index has somewhere to land without a migration.

3i. Browser-only is a soft ceiling (severity: low now, high at Raven Forest scale). The rebuild's later phases (OCR, large corpora, real Raven-Forest) likely require a Tauri 2 desktop shell with the same Vite/React frontend. Decide later; document the inflection now.

3j. The math engine reference is solid; the rebuild contract should also lock test fixtures (severity: low). Canonicalize a small Texas-baseline math fixture set as a versioned input + expected-output JSON.

4. Better Architecture (Modified, Not Replaced)

```
Shell: Vite + React + TS (today) -> Tauri 2 (when OCR lands)
Project Bundle = .landroid/ (macOS package directory)
  index.sqlite      ■■ records, FTS5, sqlite-vec, audit
  blobs/ab/cd/...  ■■ content-addressed SHA-256 files
  exports/         ■■ generated packets, never source
  manifest.json    ■■ version, schema, signing key
```

```
Domain layer (project record graph)
  Project, Party (+aliases), Document, DocumentLink,
  SourceCitation, SourceAttestation, InstrumentRecord,
  Tract, DeskMap, Lease, Unit, Wellbore,
  InterestReference, CurativeIssue, LeaseObligation,
  ImportSession, ActionPlan, ActionRecord
```

```
Action layer (typed commands -> ActionRecords -> audit_events)
  approve previews, undo snapshot, hash-chained log
```

Projections

Desk Map Runsheet Leasehold Document Packet
 Owner Index OpinionDraft ObligationCalendar
 AbstractorPackage (eDiscovery-compatible)
 AI cited-answer context

Retrieval (for AI Q&A)

SQLite FTS5 (BM25) + sqlite-vec dense + graph tools
 Reciprocal Rank Fusion -> Claude Citations API

What changed from the plan: storage moved to SQLite + content-addressed blobs; shell endpoint named as Tauri 2; LeaseObligation added; audit_events with hash chain added; OpinionDraft , ObligationCalendar , AbstractorPackage projections added; hybrid retrieval stack named.

5. Document Storage Recommendation

.landroid becomes a macOS package directory - same model as Final Cut, Logic Pro, Xcode, Bear. Three components:

- 1. Project records:** index.sqlite with FTS5 + JSON1 + R-tree extensions. Embedded vector store via sqlite-vec (Mozilla-backed, MIT/Apache 2.0).
- 2. Document bytes:** blobs/ab/cd/abcdefg...pdf content-addressed by SHA-256. Two PDFs of the same recorded instrument deduplicate to one blob.
- 3. OCR sidecars:** blobs/ab/cd/abcdefg...hocr alongside the PDF.

Export form: project.landroid.zip with BagIt-style manifest-sha256.txt and optional Ed25519 signature.

6. OCR / Text Extraction Recommendation

Local default (free, no cloud, you own the bytes): ocrmypdf wrapping Tesseract 5, with the ocrmypdf-apple plugin so Apple Vision OCR is used on macOS. Apple Vision OCR materially outperforms Tesseract on courthouse scans and is free on Apple Silicon. Output is a searchable PDF + sidecar hOCR with per-word bounding boxes and confidence scores.

Paid escalation: Mistral OCR 3 (~\$1-\$2 per 1k pages), Azure AI Document Intelligence (~\$1.50-\$10 per 1k pages), AWS Textract.

Layout-aware semantic chunking: Docling (Apache 2.0, IBM-maintained, real bbox provenance).

Side-by-side PDF-with-highlight UI: pdfjs-dist for rendering, react-pdf-highlighter for the overlay.

7. AI / RAG / Search Recommendation

Hybrid retrieval, not vector-first. Tool-use-only contract. Claude Citations as the answer primitive.

1. BM25 keyword search via SQLite FTS5.
2. Dense vector search via `sqlite-vec` .
3. Structured retrieval as tools - expose curated SQL queries as AI tools (`traverse_chain_of_title(tract_id)` , `find_burdens(lease_id)`).
4. Reciprocal Rank Fusion (RRF) combines into one ranked list.
5. Reranking (optional, deferred): Cohere Rerank 3.5 or ColBERT via RAGatouille.
6. Anthropic Contextual Retrieval before chunking.
7. Answer step: Anthropic Claude with the Citations API.

Refusal as a structural property, not a prompt: Every claim must include at least one cite. The LLM is in tool-use-only mode. If tools return nothing, the answer is "insufficient evidence" - full stop.

8. Attorney Packet / Export Recommendation

Two-format export from the same manifest. Always.

Format A: Springhill-style attorney package (today's pattern, refined): Runsheet.xlsx, TitleDocuments/, Plats/, OpinionDraft.docx, CurativeIssues.csv, Exceptions.csv, manifest.json/csv, README.txt.

Format B: eDiscovery-compatible sidecar - `eDiscoverySidecar/` with Concordance `production.dat` , Opticon `production.opt` , TEXT/ folder, exceptions.csv, README-edisco.txt.

Chain-of-custody: `manifest.json` carries SHA-256 of every file, `audit_log_digest`, Merkle root over all document hashes, optional Ed25519 signature.

9. Local Tools To Install On The User's Mac

ocrmypdf + tesseract + ocrmypdf-apple plugin + ghostscript + qpdf + poppler + mupdf + sqlite-utils + duckdb + ripgrep/fd/jq/bat + gdal + mapshaper + QGIS + Ollama (already installed) + gh + git-lfs + exiftool + Hazel + Raycast. Skip pdftcpu, Chroma/Qdrant standalone. DEVONthink Pro for personal research library only; not as LANDroid's vault.

10. Paid Services Worth Considering

- **Anthropic API (Claude)** - \$50-\$500/mo. Citations API + prompt caching is the answer-layer primitive. Single most important spend.
- **Voyage AI embeddings** - \$0.05-\$0.18 / 1M tokens.
- **Cursor or Claude Code Pro** - \$20/mo.

- **Mistral OCR 3** - ~\$1-\$2 / 1k pages.
- **Azure AI Document Intelligence** - ~\$1.50-\$10 / 1k pages.
- **Cohere Rerank 3.5** - \$1-\$3 / 1M tokens, only when needed.
- **TexasFile / CourthouseDirect subscriptions** - per-county or per-doc.
- **DEVONthink Pro** - \$199 one-time, personal research library.

A reasonable starting budget for one independent landman: ~\$100/mo (Anthropic + Voyage + Claude Code Pro), with OCR/CHD/abstractor costs ad hoc.

11. Open-Source Tools Worth Considering

- SQLite + FTS5 + JSON1 + R-tree (project DB, search, JSON, geographic indexing)
- sqlite-vec (vector search inside same file)
- DuckDB (analytical SQL)
- ocrmypdf + Tesseract 5 + Docling
- pdfjs-dist + react-pdf-highlighter
- RAGatouille (optional ColBERT reranker)
- LanceDB (alt vector store if sqlite-vec doesn't scale)

Skip: LangChain, Haystack, AutoRAG, Verba, Chroma/Qdrant/Weaviate standalone, Microsoft GraphRAG, LightRAG, txtai.

12. Specific Edits Recommended For docs/rebuild-plan.md

- 1. Add Phase 0.5 - Storage Format Change** between Phase 0 and Phase 1. Replace JSON-blob-in-IndexedDB-row workspace storage with SQLite + content-addressed blobs in a `.landroid/` macOS package directory.
- 2. Promote side-store unification to Phase 2 first task.**
- 3. Add `OpinionDraft`, `ObligationCalendar`, `AbstractorPackage` to the projection list.**
- 4. Add `LeaseObligation` and `ObligationEvent` to the record list.**
- 5. Clarify the AI Answer Contract.** Until OCR exists, AI may only cite structured records, source attestations, and explicit open issues.
- 6. Add OCR pipeline plan as Phase 2.5** (between Document Vault and ImportSession).
- 7. Add audit-log + chain-of-custody section.**
- 8. Add eDiscovery sidecar to the Runsheet Package.**
- 9. Add the Tauri 2 desktop inflection note** to "What is explicitly not approved yet."
- 10. Add a "Hybrid Retrieval Architecture" subsection.**

11. Lock test fixtures in Phase 0 (versioned input + expected-output JSON).

12. Add an `external_party_id` field to the `Party` record description in Phase 5.

13. Specific Edits Recommended For Other Docs

- **ROADMAP.md:** add side-store unification, OCR pipeline integration, hybrid retrieval architecture, Tauri 2 desktop shell decision gate, LeaseObligation projection, eDiscovery sidecar export.
- **IDEAS.md:** add macOS package directory, OpinionDraft projection, hash-chained `audit_events`, cross-project party canonical identity, Apple Vision OCR via `ocrmypdf-apple` plugin, cross-project starter records.
- **ARCHITECTURE.md:** add "Planned: Storage Format Migration", "Planned: Retrieval Architecture", "Planned: Desktop Shell" sections.
- **TESTING.md:** add storage format round-trip tests, OCR pipeline tests, audit log tamper tests, large project performance tests.
- **SECURITY.md:** add chain-of-custody section, update OCR planning, add retrieval surface subsection.

14. Risks And Migration Costs

Risk	Severity	Mitigation
JSON-blob workspace format hits Raven Forest scale	Critical	Phase 0.5 SQLite migration before Phase 1
Side-store fragmentation calcifies	High	Make unification Phase 2's first task
OCR deferred but AI cited answers promised	High	Clarify contract OR plan OCR as Phase 2.5
Browser-only ceiling hit late	Medium-High	Document Tauri 2 inflection now
Action records not durable / tamper-evident	Medium	Hash-chained <code>audit_events</code> + Merkle manifest
AI gets too clever before structured retrieval ready	Medium	Tool-use-only contract; structural refusal
Cross-jurisdiction creep into Texas math	Medium	Existing rule clear; maintain
Vendor lock-in to paid OCR/AI provider	Medium	Local <code>ocrmypdf</code> +Apple Vision default; cloud opt-in
<code>.landroid</code> v8 -> next-version migration breaks projects	Medium	Preserve v7/v8 import compat indefinitely

Risk	Severity	Mitigation
Cross-project party identity defer turns into a wall	Low-Medium	Add <code>external_party_id</code> now on <code>Party</code>
Math engine accidentally regresses during rebuild	Low (if Phase 0 honored)	Golden masters + locked math fixtures
Test suite doesn't catch Raven-Forest-scale regressions	Low-Medium	Add scale fixture + perf budget in CI

15. Final Go / No-Go Recommendation

Go, with the three amendments named in Section 1 done before Phase 2.

The plan is fundamentally sound. The architecture is the right one for the domain - closer to how Tracts.co actually models the problem than to how enterprise agreement-centric vendors do it, and more rigorous than either. The `SourceAttestation` concept is a genuine differentiator and is not present in any competitor I found.

Sequencing recommendation:

- 1. Phase 0** as written.
- 2. Phase 0.5 (new)** - Storage Format Change. SQLite + `.landroid/` package directory. New ADR.
- 3. Phase 1** as written, on the new storage foundation.
- 4. Phase 2** with first task = side-store unification.
- 5. Phase 2.5 (new)** - OCR Pipeline Integration.
- 6. Phase 3** as written.
- 7. Phase 4** as written, with `audit_events` hash chain landing here.
- 8. Phase 5** as written, with `external_party_id` field added.
- 9. Phase 6** as written; add `LeaseObligation` records here or in 5.5.
- 10. Phase 7** as written, after dedicated math design pass.

Tauri 2 desktop pivot is not a phase. It is an inflection - likely between Phase 2 and Phase 2.5, possibly later. Document in ADR; do not commit until OCR + corpus volume forces it.

Section 3 - Claude's Review 2 (Fresh Redo)

Reviewer: Claude (Anthropic, Opus 4.7). Date: 2026-05-21. Method: re-read of the same source-of-truth docs, written deliberately without re-reading Review 1 first, to surface where fresh eyes land.

1. Executive Recommendation

The plan is more right than wrong. Two things will sink it if not fixed; one thing in it isn't really wrong but is dangerously named.

The fixes:

- 1. The word "graph" in "project record graph" is misleading.** What the plan actually describes is a normalized relational schema with one graph-shaped substructure (the ownership tree). An implementer reading "graph" will reach for Neo4j or similar. Don't. The implementation is SQLite tables with foreign keys, and one of those tables happens to model a tree. Rename or qualify.
- 2. Phase 0 as written is too soft.** "Inventory every page and major workflow" sounds rigorous but doesn't say what an inventory row is. Without a frozen, executable behavior catalog, the rebuild will silently lose user-visible behaviors. This is the single most important risk in the plan and the cheapest one to mitigate.
- 3. The AI Answer Contract is half-specified and half-aspirational.** "Answer / cited sources / confidence / open issues / suggested next action" sounds professional. Three of those five fields are where hallucinations live ("confidence", "open issues", "suggested next action"). The plan needs a citation verifier that rejects any answer without a back-traceable cite to a document span, a record ID, or a deterministic tool result - and explicitly forbids the AI from inventing open issues or next actions that aren't grounded in records.

The dangerously-named thing: **"Document Vault."** Documents in the plan get a `Document` , `DocumentLink` , `SourceCitation` , `SourceAttestation` quartet. That's a normalized document model - fine. But "vault" implies storage durability, chain-of-custody, hash chains, immutable originals, derivative tracking, OCR run lineage. The plan describes the join shape, not the durability shape. If you build the joins without the durability, you have a document table, not a vault.

Beyond the three fixes: ship Phase 0 unchanged; defer almost everything else until Phase 0 lands.

2. What The Plan Gets Right

- **Rejecting event sourcing as the primary domain model.** Event sourcing is right for systems with high concurrency, audit-as-the-product, and replayable derivations. Title work is none of those. The plan correctly puts records first and actions over them.
- **Phase 0 = inventory + golden masters before any structural change.** The 627-test, 11-e2e codebase is not throwaway. Anchoring rebuild work to existing behavior is the correct discipline.

- **SourceAttestation as a first-class concept.** I haven't found another vendor naming "this chain starts from a 2018 title opinion, not from sovereignty" as a domain primitive. Most silently pretend to start from patent. The plan is more honest.
- **Documents as first-class entities, polymorphically linked.** Already 80% built in `src/storage/db.ts` at schema v9.
- **Title-opinion-as-root workflow.** This is the realistic starting state for almost every modern Texas/NM project.
- **Approval-gated AI mutations.** Already implemented and tested.
- **Dual decimal + fraction display contract.** Locking this prevents drift.
- **Federal/private math deferred behind an explicit gate.** Keeps the Texas baseline clean.
- **Math engine preserved.** LANDMAN-MATH-REFERENCE.md is the moat; the plan correctly doesn't touch it.

3. What The Plan Gets Wrong Or Under-Specifies

3a. "Project record graph" naming risk. The plan uses "graph" as a label. Parties, leases, units, wellbores, curative issues, source attestations are not graph nodes in any non-trivial sense - they're records with foreign keys. The ownership tree IS graph-shaped. **Rename to "project record schema" or "project domain records" with a callout that the ownership-tree subset is tree-shaped and gets graph algorithms.**

3b. Phase 0 is too soft. Missing: what an inventory row looks like (atomic, testable user-visible behavior); coverage criteria; a *frozen reference workspace* per demo project, exported as v8 `.landroid`, checksummed, with math outputs captured as JSON; behavior that's currently *implicit* (warning thresholds, default sort orders, autosave debouncing); performance baselines. **The Phase 0 exit gate should be: a frozen reference workspace + a behavior catalog + measured performance baselines, all checked in, all running in CI.**

3c. The AI Answer Contract is half-aspirational. Two are concrete (answer, cited sources). Three are aspirational: "Confidence / limits" (probability? heuristic? subjective?), "Open issues" (hallucination vector: the model can invent issues that aren't in the records), "Suggested next action" (similar). The fix is structural: a **CitationVerifier** module that examines every answer pre-display, extracts material claims, and rejects any claim without a back-traceable cite. "Confidence" removed OR defined as a discrete enum. "Open issues" must point to existing `CurativeIssue` records. "Suggested next action" must be a typed `ActionPlan` proposal or a navigation hint.

3d. The "Document Vault" is described as a join, not as a vault. Missing: **Immutable original** (when a PDF lands, bytes are content-addressed, re-importing is a no-op). **Document versions** (OCR'd PDF is derivative, not replacement; metadata edits are new versions). **Extraction runs as lineage records** (engine, version, parameters, timestamps, output artifacts). **Hash-chained audit log** (even single-user, "I can prove I didn't tamper with this packet" is a feature).

3e. Workspace storage is the iceberg. Workspace lives as a single JSON string in one Dexie row, autosaved on changes. For Raven Forest, this row reaches tens of megabytes. Three options, ordered cheapest-to-most-aggressive:

1. **Shard the workspace** into per-table Dexie rows. Keeps the browser-first posture. Maybe 1-2 weeks.
2. **Move to SQLite WASM in OPFS.**
3. **Tauri 2 desktop shell with native SQLite.**

Option 1 is the right Phase 0.5 amendment - cheapest, lowest risk, doesn't commit the rebuild to a shell pivot it might not need.

3f. Math engine input surface isn't insulated. The rebuild *will* change how nodes, leases, units, ORRIs, NPRIs are represented. If the math engine reads from new representations directly, the rebuild can quietly break math semantics. **Name an `MathInputView` projection that the math engine consumes** - a stable shape that doesn't change as records reorganize.

3g. Side-store fragmentation. `owners` , `ownerDocs` , `mapAssets` , `researchImports` , `documents` are five tables that all store assets in different shapes. **The unification has to happen before the vault is meaningfully "the" vault.**

3h. Things the plan leaves unaddressed entirely:

- **Multi-tab safety.** Two LANDroid tabs against the same workspace. Today, autosave race conditions are possible.
- **Multi-machine sync.** Single user, two Macs - common. Plan implies manual `.landroid` import/export.
- **Print fidelity.** Landmen still print runsheets and flowcharts.
- **In-flight project migration.** During rebuild, what happens to a workspace mid-flight?
- **Telemetry for catching AI failures in production.** With hosted deployment live, AI wrong answers happen silently.

4. Better Architecture

Records (relational schema, one tree-shaped subset)

- Project, Party (+aliases), Tract, DeskMap, Lease, Unit, Wellbore, InterestReference, CurativeIssue, ImportSession, ActionPlan, ActionRecord
- Document (immutable original, content-hashed)
- DocumentVersion (metadata edits, derivative tracking)
- DocumentLink (polymorphic join - already implemented)
- ExtractionRun (OCR/extraction lineage)
- SourceCitation (any fact / doc page/span)
- SourceAttestation (root-of-chain provenance)

Action Layer (typed commands, approval previews, undo, audit chain)

- ActionPlan (typed proposal, dry-run-able)
- ActionRecord (immutable post-approval record)
- AuditEvent (append-only, prev_hash-chained)
- CitationVerifier (rejects answers without back-traceable cites)

Math Layer (consumes a stable projection, not raw records)

- MathInputView -> math-engine.ts (unchanged semantics)

Projections (read-only views)

- Desk Map · Runsheet · Leasehold · Documents · Owners · Curative
- Packet export · AI context

Storage

- Phase 0.5: shard workspace into per-row Dexie tables
- Later: SQLite WASM in OPFS, when query needs justify it
- Eventually: Tauri 2 native shell, when OCR/corpus forces it

Differences from the plan: added `DocumentVersion`, `ExtractionRun`, `AuditEvent` (hash-chained), `CitationVerifier`, `MathInputView` insulator; staged storage migration, not a single jump.

5. Document Storage

Today: shard the workspace blob within IndexedDB. One row per node, one per lease, one per ORRI. Eliminates the big-blob autosave problem at minimal cost.

Next: introduce a content-addressed blob layer. SHA-256-keyed slot. Already half-done; just enforce it.

Later: SQLite WASM in OPFS when the query story makes Dexie painful. OPFS isn't user-visible in Finder, which is a real problem if the user expects Time Machine backup or emailing a project. That's the trigger for:

Eventually: Tauri 2 desktop shell with native SQLite, native filesystem, native OCR process spawning. ADR-able when triggered.

6. OCR / Text Extraction

Three-step recommendation:

- 1. Don't promise document-citing AI before OCR exists.** Today, AI may only cite structured records, source attestations, curative issues.
- 2. When ready, the local pipeline is** `ocrmypdf` **+** `ocrmypdf-apple` **plugin** (Apple Vision OCR is materially better than Tesseract on courthouse scans and is free on macOS). Output: searchable PDF + hOCR sidecar with per-word bounding boxes.
- 3. Cloud OCR is per-document opt-in**, not a default. Mistral OCR 3 and Azure Document Intelligence are the right paid escalations.

Citation schema: `{document_id, version_id, extraction_run_id, page, bbox, char_span, text, confidence}`. Drive a `pdfjs-dist` viewer with `react-pdf-highlighter` overlays.

7. AI / RAG / Search

Search is hybrid, not vector. Title work is full of exact-string queries (instrument numbers, party names, volume/page). Vector embeddings smooth over precisely the tokens that matter.

- **SQLite FTS5** for keyword/exact search.
- `sqlite-vec` (inside same DB file) for embedding-based search.
- **Structured tool calls** for graph traversal.
- **Reciprocal Rank Fusion** combiner.

Answers go through Claude with the Citations API. Returns guaranteed-valid `page_location` + `cited_text` spans. The `CitationVerifier` module rejects answers that fail to back-trace.

Tool-use-only contract. The AI's general knowledge is off. Every claim must come from a tool result or a cited document span.

Skip: LangChain, Haystack, heavy LlamaIndex stack, Microsoft GraphRAG framework, third-party vector DBs.

8. Attorney Packet

The plan's Springhill-style format is correct for the human deliverable. Two amendments:

- 1. Add a deterministic sidecar.** `manifest.json` + per-doc text files + `production.dat` (Concordance-style) + `production.opt` (Opticon page index) + `checksums-sha256.txt` .
- 2. Sign the manifest.** Merkle root over all document hashes + Ed25519 signature. Defensible chain-of-custody.

9. Local Tools On The Mac

The ones that matter, ranked by leverage:

- `ocrmypdf` + `ocrmypdf-apple` plugin (local OCR with Apple Vision) - free
- `tesseract` , `ghostscript` , `qpdf` (OCR engine + PDF tooling) - free
- `poppler` (`pdftotext` , `pdfimages` , `pdftk`) - free
- `mupdf` / `mutool` (fastest PDF rendering) - free
- `sqlite` , `sqlite-utils` , `duckdb` - free
- `ripgrep` , `fd` , `jq` - free
- `gdal` , `mapshaper` (shapefile/GeoJSON ops) - free
- `exiftool` - free
- Ollama (already installed) - free
- Hazel (auto-route PDFs to project intake) - \$42 one-time

Skip: standalone vector DBs, LangChain dependencies, AI document assistant SaaS that isn't doing something you can't do locally.

DEVONthink Pro: personal research library only.

10. Paid Services

- **Anthropic Claude API** - primary AI provider. Citations API. ~\$50-500/mo.
- **Voyage AI embeddings** - pair with Anthropic.
- **Mistral OCR 3 OR Azure AI Document Intelligence** - paid OCR fallback.
- **Claude Code Pro / Cursor** - dev velocity.

Don't subscribe to: standalone vector DB SaaS, Pinecone, LangChain Cloud, Reducto unless tables become a bottleneck, Google Document AI.

11. Open-Source Tools

ARE the stack (not "consider for later"):

- SQLite (FTS5, JSON1, R-tree) + `sqlite-vec` + DuckDB
- `ocrmypdf` + Tesseract + `ocrmypdf-apple`
- `pdfjs-dist` + `react-pdf-highlighter`
- Existing: Vite, React, TS, Decimal.js, Zod, Dexie (until SQLite migration), Vitest, Playwright

12. Specific Edits To docs/rebuild-plan.md

1. **Rename "project record graph" to "project record schema."** Five words; saves a future architecture argument.
2. **Tighten the Phase 0 exit gate.** Require: frozen reference workspace; behavior catalog; measured performance baselines; CI-running golden masters.
3. **Extend the AI Answer Contract to a structural one.** CitationVerifier rejects uncited claims. `confidence` becomes a discrete enum. `open issues` must point to existing `CurativeIssue` records. `suggested next action` must be a typed `ActionPlan` or navigation hint.
4. **Extend Document Vault into a vault.** Add `DocumentVersion`, `ExtractionRun`, content-hash immutability, `AuditEvent` hash chain, Merkle-rooted signed export manifest.
5. **Add Phase 0.5: workspace sharding.** Move workspace from JSON-blob-in-a-row to per-table Dexie rows.
6. **Add an explicit `MathInputView` projection** decoupling math from underlying record shape.
7. **Promote side-store unification to Phase 2's first task.**

8. Add a **"Print Fidelity" rebuild contract** to the Non-Negotiable Contract section.
9. Add an **"In-Flight Project Migration" rebuild contract**.
10. Add a **"Multi-Tab Detection" item** to the Open Risks list.
11. Add a **"Telemetry For AI Failures" item** to roadmap-level concerns.
12. Reframe Tauri 2 desktop as a **documented future inflection**, not a phase.

13. Edits To Other Docs

- **ROADMAP.md**: elevate workspace sharding and Document Vault durability to *Now*. OCR pipeline as *Next*. Tauri desktop is *Later*.
- **IDEAS.md**: move tokenized search and saved packet selections from *Parked* to *Promising*.
- **ARCHITECTURE.md**: add "Math Input View" section. Add "Citation Verifier" section.
- **TESTING.md**: add explicit golden-master fixture format (workspace `.landroid` snapshot + expected math JSON), scale fixture, multi-tab race detection tests, AI citation verifier tests.
- **SECURITY.md**: add chain-of-custody section (append-only audit, hash chain, signed manifest). Add OCR data residency rules.

14. Risks And Migration Costs

- **Phase 0 too soft -> silent regressions**. Whole rebuild loses credibility.
- **AI contract aspirational -> hallucinated cites**. Title work is wrong; trust gone.
- **Workspace JSON-blob hits scale**. Raven Forest project crashes or stalls.
- **Document Vault is just joins -> no chain-of-custody**. Packet defensibility weak.
- **Math input not insulated -> rebuild changes math semantics**. Quiet incorrect math.
- **Side-store fragmentation persists -> vault is incomplete**. AI retrieval blind spots.
- **Cross-project party identity never lands -> cross-project search impossible**.
- **Multi-tab races -> silent corruption**.
- **Cloud AI sees sensitive title content without per-project policy**. PII exposure.
- **Print fidelity quietly breaks**. Real workflows fail post-rebuild.

15. Final Go / No-Go

Go on Phase 0 as written, with the strengthened exit gate.

No-go on Phase 1+ implementation until the plan is amended with: the renamed schema framing, the structural AI contract, the durability-bearing Document Vault, the math input insulator, the workspace sharding Phase 0.5, and the named rebuild contracts (print fidelity, in-flight migration).

Go on the architecture itself after amendments. Records + actions + projections + a citation-verified AI layer is right. The plan just needs to mean what it says.

Section 4 - Side-By-Side Comparison

*Cross-comparison of Codex's review (Section 1) and Claude's two reviews (Sections 2 and 3). Author: Claude.
Method: read all three reviews; identify agreements, disagreements, and items unique to each.*

A. Agreements (where all three reviews land in the same place)

These are areas where two or three independent reviews converge. Treat convergence as a strong signal:

- The plan's direction is right; adopt with amendments.
- Phase 0 = Go; preserve inventory + golden masters discipline.
- Project record + action layer + projections is the correct shape.
- Math engine preservation is non-negotiable.
- `SourceAttestation` is a real differentiator vs every competitor surveyed.
- Side-store unification is required before the Document Vault is meaningful.
- AI must structurally refuse uncited answers; not rely on prompts alone.
- Hybrid retrieval (BM25 + vector + graph + deterministic math tools) beats vector-only.
- Claude Citations API is the preferred answer primitive.
- `ocrmypdf` + Tesseract is the local OCR pipeline default.
- Document blobs do not belong inside JSON workspace or event logs.
- Federal/private math defer is correct.
- Don't adopt LangChain/Haystack/standalone vector DBs.
- Workspace storage at Raven Forest scale is a real risk.
- Concrete `.landroid` package layout is required before the format work begins.

B. Disagreements and emphasis differences

These are places where the two reviewers (or the two Claude reviews) reach different conclusions. Each is named with a recommended resolution.

B.1 - Storage migration aggressiveness

- **Codex:** Keep IndexedDB/Dexie short-term; target a `ProjectName.landroidpkg/` folder. Hedge between `project.json` and `project.sqlite`.
- **Claude Review 1:** Aggressive: SQLite + content-addressed blobs in a macOS package directory, as a new Phase 0.5.

- **Claude Review 2:** Measured: shard the workspace blob within Dexie first; SQLite WASM as a later step; macOS package directory as eventual; Tauri as eventually.

Resolution: Review 2's staged approach is the right one. Phase 0.5 should be "shard the workspace into per-row Dexie tables" (cheapest meaningful win). SQLite migration and macOS package directory are documented future trajectories, not current commitments. The cost of doing the cheap thing first is low; the cost of locking into SQLite before profiling shows it's needed is higher.

B.2 - Tauri 2 desktop pivot

- **Codex:** Doesn't mention it.
- **Claude Review 1:** Documented inflection in ARCHITECTURE.md and ADR-candidate.
- **Claude Review 2:** Same direction, less prominent - explicitly downgraded to "later."

Resolution: Review 1's framing is correct. Document the inflection now in ARCHITECTURE.md so the implementer knows OCR + corpus volume will force the pivot. Do not commit to it until triggered.

B.3 - AI Answer Contract specificity

- **Codex:** "Citation verifier that refuses answers unless each material claim maps to a stored SourceCitation or deterministic calculation result." Named the concept.
- **Claude Review 1:** "No claim without a cite" as a structural property.
- **Claude Review 2:** Most specific - CitationVerifier as a separate named module; confidence reduced to discrete enum; open issues must point to existing CurativeIssue records; suggested next action must be a typed ActionPlan or navigation hint.

Resolution: Review 2's framing is the most concrete and correct. Adopt the structural refusal as Review 2 describes it.

B.4 - Missing projections

- **Codex:** Doesn't raise these.
- **Claude Review 1:** Names three: OpinionDraft , ObligationCalendar , AbstractorPackage .
- **Claude Review 2:** Misses all three (regression in fresh pass).

Resolution: Review 1 is right. Add all three projections. Every upstream land vendor surveyed produces an opinion-shaped output and an obligation calendar. The competitive case for LANDroid requires them.

B.5 - LeaseObligation as a record

- **Codex:** Doesn't raise.
- **Claude Review 1:** Names it explicitly. Every upstream land vendor has one.

- **Claude Review 2:** Misses it (regression).

Resolution: Review 1 is right. Add `LeaseObligation` and `ObligationEvent` to the record list. Missing a lease deadline is the failure mode that costs the landman real money.

B.6 - Cross-project party identity

- **Codex:** Doesn't address.
- **Claude Review 1:** Adds `external_party_id` field hook on `Party` for future cross-project canonical identity.
- **Claude Review 2:** Mentions in risks; less prominent.

Resolution: Review 1 is right. Add the field now. Cheap; preserves the option.

B.7 - Phase 0 rigor

- **Codex:** Endorses Phase 0 but doesn't specify exit gate.
- **Claude Review 1:** Mentions briefly; not the headline.
- **Claude Review 2:** Promotes Phase 0 strengthening to top-3 risk; specifies exit-gate criteria (frozen reference workspace, behavior catalog, performance baselines, CI-running golden masters).

Resolution: Review 2 is right. Phase 0 as written is too soft. Without the strengthened exit gate, the rebuild will silently lose user-visible behaviors.

B.8 - "Project record graph" naming

- **Codex:** Doesn't flag.
- **Claude Review 1:** Doesn't flag.
- **Claude Review 2:** Flags as a real risk - will pull the implementer toward graph DBs unnecessarily.

Resolution: Review 2 is right. Five-word rename (to "project record schema" or "project domain records") prevents a future architecture argument.

B.9 - `MathInputView` insulator

- **Codex:** Doesn't raise.
- **Claude Review 1:** Doesn't raise.
- **Claude Review 2:** New - names an explicit projection that the math engine consumes so the rebuild can't quietly change math semantics.

Resolution: Review 2 is right. Add the insulator. Cheap; eliminates a real risk.

B.10 - Multi-tab safety, in-flight migration, print fidelity, telemetry

- **Codex:** Doesn't address.
- **Claude Review 1:** Doesn't address.
- **Claude Review 2:** All four named as missing items.

Resolution: Review 2 is right. Add all four as rebuild contracts.

B.11 - eDiscovery sidecar

- **Codex:** Hand-rolls a `loadfile.csv` ("DocID, native path, text path, document date, parties, instrument number, volume/page, tract, source type, hash"). Good but doesn't import into Relativity/Everlaw/DISCO directly.
- **Claude Review 1:** Prominent recommendation with Concordance `.DAT` + Opticon `.OPT` / `.LFP` + family relationships (`BegAttach` / `EndAttach`). Importable into any litigation system.
- **Claude Review 2:** Mentioned but lighter version.

Resolution: Review 1 is right. Adopt Concordance/Opticon format. Costs almost nothing if data shape is right; makes LANDroid packages instantly importable into any eDiscovery system.

B.12 - `DocumentVersion` and `ExtractionRun` as formal first-class records

- **Codex:** Schema is sharpest - `ExtractionRun` with explicit fields (documentId, documentVersionId, engine, engineVersion, startedAt, completedAt, status, confidenceSummary, parameters, outputArtifactIds). `DocumentVersion` named.
- **Claude Review 1:** Half-named in OCR section; not promoted to vault contract.
- **Claude Review 2:** Promoted to vault contract, named as new records.

Resolution: Codex is right. Use Codex's `ExtractionRun` schema verbatim. Add `DocumentVersion` . Both belong in Phase 1's record list.

B.13 - Action layer sequencing

- **Codex:** Phase 3 already needs typed `ActionPlan` dry runs. Phase 4 makes records canonical. The schema must exist when Phase 3 starts.
- **Claude Review 1:** Didn't address.
- **Claude Review 2:** Didn't address.

Resolution: Codex is right. Fold the `ActionPlan` schema definition into Phase 1's record list; Phase 3 consumes it; Phase 4 makes it the single mutation path.

B.14 - Backend adapter boundaries

- **Codex:** Frames as a decision gate - adapter shapes for blob storage, OCR jobs, AI inference need boundaries now so a backend can be added without storage rewrite.
- **Claude Review 1:** Same family of concern, but framed as Tauri 2 desktop shell inflection.
- **Claude Review 2:** Same as Review 1.

Resolution: Codex is right that backend-vs-desktop are separate decisions. Adopt both: backend adapter shape (Codex) AND Tauri desktop inflection (Claude Review 1). They address different concerns.

B.15 - Vendor competitive intelligence

- **Codex:** Names Enverus Courthouse, Tracts, Quorum, NetDocuments, iManage as benchmarks.
- **Claude Review 1:** Deeper - Tracts.co at \$250/day or \$2,500/month/seat as the price ceiling for an independent landman; the only direct competitor at this price point.
- **Claude Review 2:** Omitted.

Resolution: Review 1 is right. The Tracts.co price ceiling is product-positioning intelligence the user should keep. Independent landman tool can stay under that ceiling and win.

B.16 - Mac tool list

- **Codex:** Conservative list of OSS Mac dev tools.
- **Claude Review 1:** Longer - includes Raycast, DEVONthink (as personal library only), Hazel, the `ocrmypdf-apple` plugin.
- **Claude Review 2:** Shorter than Review 1, but still includes Hazel and `ocrmypdf-apple`.

Resolution: Review 1's list is the right scope. Hazel + `ocrmypdf-apple` are real workflow wins on Mac. DEVONthink Pro for personal research library (statutes, AG opinions, geology references) - not LANDroid's vault.

C. Consolidated amendment list (Codex + Claude Review 1 + Claude Review 2)

In priority order - what would need to land in `docs/rebuild-plan.md` before Phase 1 implementation starts:

Tier 1 - Must land before Phase 1 starts

1. **Phase 0 strengthened exit gate** (Review 2) - frozen reference workspace + behavior catalog + measured performance baselines + CI-running golden masters.
2. **Rename "project record graph" to "project record schema"** (Review 2) - with a callout that the ownership-tree subset is graph-shaped.

3. Phase 0.5: workspace sharding within IndexedDB (Review 2) - per-row Dexie tables; eliminates JSON-blob autosave problem.

4. Evidence Vault Contract (Codex + Review 1 + Review 2) - extend Document Vault with `DocumentVersion`, content-hash immutability, `ExtractionRun`, `AuditEvent` hash chain, derivative tracking.

5. `ActionPlan` / `ActionRecord` schema in Phase 1 (Codex) - so Phase 3 `ImportSession` can consume; Phase 4 still owns canonicalization.

6. Citation Verifier as a named structural module (Codex + Review 2) - examines every answer pre-display, rejects unverified claims. `confidence` discrete enum. open issues point to existing `CurativeIssue` records. suggested next action typed `ActionPlan` or navigation hint.

7. `MathInputView` projection (Review 2) - insulator between domain records and `math-engine.ts`.

8. AI Answer Contract clarified (Review 1 + Review 2) - pre-OCR, AI cites only structured records, source attestations, explicit open issues; document-text citations off-limits.

9. Hybrid retrieval contract (Codex + Review 1 + Review 2) - FTS5 (BM25) + `sqlite-vec` + graph tools + deterministic math tools + Anthropic Citations API + `CitationVerifier` + tool-use-only.

Tier 2 - Must land in Phase 2 or 3

10. Side-store unification at top of Phase 2 (Codex + Review 1) - collapse `ownerDocs` / `mapAssets` (PDF subset) / `researchImports` (file subset) into unified `documents`.

11. OCR pipeline as Phase 2.5 (Review 1 + Review 2) - `ocrmypdf` + `ocrmypdf-apple` plugin default; Mistral OCR 3 / Azure Document Intelligence as paid escalation. `ExtractionRun` (lineage), `SourceCitation` records.

12. `OpinionDraft`, `ObligationCalendar`, `AbstractorPackage` projections (Review 1).

13. `LeaseObligation` + `ObligationEvent` records (Review 1).

14. eDiscovery sidecar (Review 1 + Review 2) - Concordance `.DAT` + Opticon `.OPT` deterministically generated. Family relationships via `BegAttach` / `EndAttach`.

15. Print fidelity rebuild contract (Review 2) - runsheets and flowcharts print correctly post-rebuild.

Tier 3 - Decision gates, not features

16. Backend adapter boundary (Codex) - blob storage, OCR job dispatch, AI inference adapter shapes defined now.

17. Tauri 2 desktop shell inflection (Review 1) - documented planned inflection; ADR-able when triggered.

Tier 4 - Chain-of-custody and identity hooks

18. Hash-chained `AuditEvent` table (Review 1 + Review 2) - append-only, `prev_hash` per row.

19. Merkle-rooted signed manifest (Review 1) - export includes SHA-256 + audit-log digest, optional Ed25519 signature.

- 20. `external_party_id` field on `Party` (Review 1).
- 21. Versioned Texas-math fixture set (Review 1).
- 22. In-flight project migration rebuild contract (Review 2) - every phase preserves loadability of pre-rebuild workspaces.
- 23. Multi-tab detection (Review 2) - workspace lock token, last-write-wins with conflict surfacing.
- 24. Telemetry for AI failures (Review 2) - verifier rejections, proposal rejections, undo counts.

D. Net Take

The three reviews land in the same architectural place: project record schema + evidence-grade document vault + typed action layer + citation-verified AI. The differences cluster in three places:

- 1. **How aggressive the storage change should be.** Review 1 was over-aggressive (SQLite + macOS package directory now); Codex hedged; Review 2's "shard first" is right.
- 2. **Whether the AI contract needs structural reshaping.** Review 2 + Codex are most concrete; adopt their framing.
- 3. **Which projections and records to add.** Review 1 was most complete (`OpinionDraft` , `ObligationCalendar` , `LeaseObligation` , `AbstractorPackage` , eDiscovery family relationships, `external_party_id`); Codex was sharpest on `ExtractionRun` schema; Review 2 was sharpest on `MathInputView` .

The honest combined recommendation:

- Take Review 2's risk framing (naming risk, soft Phase 0, AI hallucination slots, math input insulation, multi-tab, print fidelity, in-flight migration, telemetry).
- Take Review 1's projections and records (`OpinionDraft` , `ObligationCalendar` , `LeaseObligation` , `AbstractorPackage` , Concordance/Opticon eDiscovery sidecar with family relationships, `external_party_id` , vendor competitive intelligence).
- Take Codex's `ExtractionRun` schema and action-layer-in-Phase-1 sequencing.
- Take Review 2's measured storage trajectory (shard first, SQLite later, package directory eventually, Tauri eventually).
- Take Codex's backend adapter shape definition AND Review 1's Tauri inflection (these are separate concerns).

Section 5 - Recommended Next Steps

Author: Claude. Subject: how to actually use this document to produce an amended rebuild plan, and how the follow-up ultra audits should be run.

What this document is for

This PDF exists to give the next reviewer (Codex, in this case) a complete, structured view of two independent architectural reviews of `docs/rebuild-plan.md`. The expected use is:

1. Codex reads Section 1 (its own review) for orientation.
2. Codex reads Section 2 (Claude Review 1) and Section 3 (Claude Review 2) for the other perspective.
3. Codex reads Section 4 (the comparison) to see where the reviews agree, disagree, and which side was identified as correct on each disagreement.
4. Codex uses Section 4's consolidated amendment list (Tier 1 through Tier 4) as the basis for an amended `docs/rebuild-plan.md`.

Recommended sequence

Step 1. Amend the plan, do not rewrite it.

The current `docs/rebuild-plan.md` is structurally right. The amendments in Section 4 ("Consolidated amendment list") are scoped edits, not a new plan. Codex should:

- Edit `docs/rebuild-plan.md` directly, incorporating Tier 1 amendments first.
- Update `ROADMAP.md`, `IDEAS.md`, `ARCHITECTURE.md`, `TESTING.md`, and `SECURITY.md` to match the amended plan.
- Add new ADRs where a long-lived boundary changes (proposed candidates: ADR 0005 Storage Format Trajectory; ADR 0006 AI Citation Contract; ADR 0007 Action Layer Schema).
- Produce a diff for the user to review before committing.

The amended plan and its supporting docs become the artifact under audit in the next step.

Step 2. Run two ultra audits. Adversarial, not collaborative.

The two audits should NOT help create the plan. The plan-creation step is Step 1, owned by Codex with the user. The audits' job is to find what the amended plan still gets wrong.

Two audits, deliberately different in lens:

Audit A - Implementation / Technical

Focus: can this be built as described? Concerns include:

- Code-level dependencies and sequencing across phases.
- Type safety and schema migration paths.
- Test coverage adequacy for each phase exit gate.
- Performance against Raven Forest scale fixtures.
- Security and threat-model changes (especially Tauri-pivot threats, hash-chained audit log integrity, hosted AI proxy posture).
- Failure modes during in-flight migration.
- Whether each phase ends with a working app.

Audit B - Product / Domain

Focus: does this actually solve the landman's problem? Concerns include:

- Workflow fidelity vs current Desk Map / Runsheet / Leasehold / Flowchart / Documents / Curative / Maps / Research / Federal Leasing flows.
- Domain correctness for Texas math (NPRI fixed vs floating, ORRI bases, lease coverage clipping, NRI bases).
- Output quality for attorney packets, opinion drafts, abstractor handoffs.
- AI behavior in realistic title workflows (multi-step row imports, owner creation, lease attachment, curative creation).
- Whether `SourceAttestation` , `CurativeIssue` , `OpinionDraft` , `ObligationCalendar` shape the deliverables a landman actually needs.
- Whether the amended plan still distinguishes leasehold-side records from mineral-side records correctly.

The audits should not see each other's reports until both have finished.

Step 3. Apply Audit A and Audit B amendments. Optionally run a third audit.

After both audits return, the user should:

- Apply their amendments to `docs/rebuild-plan.md` directly.
- Where Audit A and Audit B disagree, the user adjudicates.
- Only if a non-trivial disagreement appears, optionally run a third audit scoped narrowly to the disagreement.

Step 4. Start Phase 0.

Phase 0 implementation (inventory + golden masters + frozen reference workspace + behavior catalog + measured performance baselines) can start as soon as the plan is amended and audited. Phase 1+ waits for

Phase 0 exit gate.

Should the ultra audit instruct the reviewer to help write the plan?

No. Here is the reasoning.

The plan-creation step (Step 1 above) is owned by Codex working with the user. Its job is to take the consolidated amendment list in Section 4 and produce an amended `docs/rebuild-plan.md`. That is collaborative work between the user and one reviewer (Codex). It is not auditable.

The audit step (Step 2) is adversarial. Each audit's job is to find what the amended plan still gets wrong. If an audit is also writing the plan, it has nothing left to attack - it has author's blindness over its own work.

The most thorough sequence is: one reviewer writes the amended plan; two independent audits attack it.

If a single reviewer both writes and audits the plan, you get one round of work, but you lose the adversarial verification layer that catches the things the writer missed. The two reviews in this PDF demonstrate why that matters - even a single reviewer (Claude) writing two passes over the same plan caught different things in each pass.

How to structure each ultra audit prompt

Each audit prompt should include:

- The current state: branch, current `docs/rebuild-plan.md`, current `ROADMAP.md` / `IDEAS.md` / `ARCHITECTURE.md` / `TESTING.md` / `SECURITY.md`.
- The amended plan: the diff or full file produced in Step 1.
- The consolidated amendment list from Section 4 of this PDF (so the audit knows what amendments were intended).
- The audit lens (A or B as defined above).
- The audit deliverable: a written 15-section review in the same shape as Section 1, 2, or 3 of this PDF. Same heading numbers; same final go/no-go.
- Explicit instructions to look for what the amended plan still gets wrong, NOT what is already addressed.
- Instruction not to read the other audit until done.

Keep each audit's prompt under 4,000 words. Longer prompts dilute focus.

How many audits is enough?

- One audit: minimum. Misses things.

- Two audits (one technical, one domain): the recommended sequence. High coverage; manageable cost; adversarial.
- Three audits: only if the first two disagree on something material that can't be resolved by re-reading the plan.
- More than three audits: diminishing returns; better to start Phase 0 and let real implementation surface what's wrong.

For LANDroid specifically, two audits is the right number.

Cost considerations

Two ultra audits + one amended plan from Codex is one of the largest analytical efforts the LANDroid project has invested in. That investment is justified because:

- The decisions made now affect every future phase.
- The math engine and approval-gated AI are real competitive assets; preserving them through a rebuild is non-trivial.
- The user's actual workload (Raven Forest scale, federal/private prep, attorney-grade output) demands a defensible architecture before implementation.

The investment is not justified for plan changes that touch only one phase. For those, a single reviewer pass is enough.

What to do if the audits surface a disagreement with this PDF

The reviews in this PDF are dated 2026-05-21. If a later audit finds a structural problem the reviews missed, treat the audit as authoritative for that problem. The reviews in this PDF are not a freeze; they are inputs to a better plan.

Final summary

- Use this PDF as the structured input for an amended `docs/rebuild-plan.md`.
- Codex amends the plan; do not have the audits write it.
- Run two ultra audits, one technical and one domain, against the amended plan. Adversarial, not collaborative.
- Apply audit amendments; optionally run one narrow third audit only if the first two disagree materially.
- Start Phase 0 as soon as the plan is amended and audited.

The plan will be ready when it has survived: two reviews (in this PDF), one amendment pass (Codex), and two adversarial audits. That is the level of rigor the rebuild deserves.